TAMIL NADU SMART TRANSPORT AI • TRANSITAI

Complete Technology Stack & System Architecture

Executive Technical Document

PLATFORM EXECUTIVE OVERVIEW

Tamil Nadu Smart Transport AI (TransitAI) is a full-featured, intelligent transportation management and predictive telematics platform. Built for the Tamil Nadu State Transport Corporation (TNSTC) highway network, it seamlessly integrates shortest-path highway routing, Machine Learning arrival delay forecasting, live GPS vehicle telemetry, automated satellite weather monitoring, and modern multi-theme UI engineering.

1. Core Infrastructure & Deployment

- **Deployment Mode:** Zero-Server Browser Standalone + Flask API
- **Primary Language:** JavaScript (ES6+) & Python 3.10+
- **Runtime Environment:** Modern Browsers & Python CPython 3.x
- **Single-File App:** index.html (330 KB standalone architecture)
- **Microservices App:** Flask RESTful Server (app.py)
- **Data Serialized:** JSON, SQLite3, Joblib, CSV Tables
- **Package Ecosystem:** Zero external npm build deps (Vanilla CSS/JS)

2. Geographic & Highway Coverage

- **Revenue Districts:** All 38 Tamil Nadu Revenue Districts
- **Transit Corridors:** NH 45 (GST), NH 44, NH 544, NH 83
- **Elevation Index:** Ghat Roads (Nilgiris/Ooty, Kodaikanal)
- **Major Hubs:** Chennai CMBT, Trichy Central, Salem, Madurai
- **RTO Series:** Official TN RTO Database (TN 01 to TN 83)
- **Fleet Livery:** SETC Super Express, Ultra Deluxe, AC Sleeper
- **Coordinate Bounds:** Lat: 8.08°N - 13.12°N | Lon: 76.69°E - 80.27°E

3. High-Level Modular Architecture

[Layer 1: Presentation & HUD Cockpit]

- Modern Spendorio UI Glassmorphism card system with multi-theme switcher (6 themes).
- Moving SETC Super Express coach on asphalt highway with scrolling broken yellow lines.
- Live synchronized IST Digital Transit Clock with real-time green beacon heartbeat pulse.

[Layer 2: Geospatial & Route Graph Engine]

- Dijkstra Highway Road Network Graph connecting all 38 districts with dynamic milestones.
- Leaflet.js interactive maps with CartoDB dark/light tile layers and SVG vehicle markers.

[Layer 3: Machine Learning & Telematics Engine]

- Scikit-learn Regression pipeline predicting arrival delay in minutes (97.8% accuracy).
- Multi-sensor telemetry: Speedometer (0-120 km/h), odometer, engine health, coolant temperature.

[Layer 4: Storage & Background Database Engine]

- LocalStorage database managing registered user directory, access permissions, and tickets.
- Real-time username & email availability verification on signup; verified email password reset.

Frontend Engineering & Client Technologies

UX, Audio, Maps & Canvas

HTML5 & Modern CSS3 Engine

- **Layout System:** CSS3 Flexbox, CSS Grid, Multi-layer Z-index
- **Visual Design:** Glassmorphism, frosted backdrop blur (20px)
- **Hardware Accel:** GPU transform: translate3d, will-change
- **Animations:** @keyframes: scrollLoginRoad, driveBusAcrossScreen
- **Wheel Physics:** @keyframes spinWheel with alloy rim and tyre
- **Responsive Design:** Mobile bottom nav (<=768px) & desktop sidebar
- **Light/Dark Themes:** 6 consistent themes via body class toggle
- **Accessibility:** Semantic forms, ARIA roles, high-contrast labels

Geospatial Mapping & Analytics

- **Mapping Library:** Leaflet.js v1.9.4 (Vector Tile Engine)
- **Map Providers:** CartoDB Dark Matter / OSM CartoDB Positron
- **Polylines:** Dynamic highway path rendering with glows
- **Marker Cluster:** SVG custom bus pin with direction arrows
- **Analytics Engine:** Chart.js v4.4 (HTML5 Canvas 2D)
- **Hourly Surge:** Smoothed bezier spline line charts
- **Fleet Health:** Doughnut charts showing on-time performance
- **Radar Cockpit:** Circular SVG SVG stroke-dasharray gauges

Web Audio API Air Horn Synthesizer

The moving SETC Express bus on the login screen features a real-time synthesized air horn with zero external MP3s:

- AudioContext Engine: HTML5 Web Audio API.
- Dual Oscillators: Simultaneous sawtooth wave generators.
- Musical Frequencies: 370.0 Hz (F#4) + 466.1 Hz (A#4).
- Authentic Harmonics: Perfectly matches dual-pitch Indian express highway coach pneumatic air horns.
- Exponential Decay: GainNode ramps volume down smoothly over 0.6 seconds (exponentialRampToValueAtTime).
- Interactive Trigger: Click or tap moving coach to blast horn, randomize highway speed, and show telemetry toast.
- Autoplay Safe: Graceful fallback for strict browser rules.

6-Theme Handcrafted Color Palettes

The application features a complete multi-theme system:

1. Spendorio Sunset Bright [Daylight]: Crisp white glass cards, vivid coral accents, optimized for sunlight visibility.
2. Cosmic Cyber Violet [Night]: Deep midnight violet with neon magenta glowing borders and synthwave futuristic vibe.
3. Nilgiris Emerald Eco [Eco]: Alpine forest green and mint accents inspired by Ooty and eco-friendly EV transit.
4. Marina Bay Ocean [Azure]: Coastal navy and cyan azure reflecting Chennai's Marina Beach and Bay of Bengal.
5. Royal Amber Highway [Gold]: Warm espresso and golden sodium streetlight tones for luxury night coaches.
6. Cyberpunk Synthwave [Neon]: High-contrast electric purple and laser cyan accents.

Algorithms, Machine Learning & Data Science

Dijkstra Highway Routing Algorithm

- **Graph Model:** Weighted Adjacency Graph of 38 Distinct Nodes
- **Cost Metric:** Highway Kilometers & Speed Limits
- **Priority Queue:** Min-Heap path relaxation implementation
- **Transit Hubs:** Automatic detection of central junctions
- **Highway Corridors:** Tracks GST Road, Salem Bypass, NH-44
- **Distance Matrix:** Precomputed haversine & road coefficients
- **Milestone Tracker:** Calculates km checkpoints every 25 km
- **Route Interpolation:** Dynamic latitude/longitude step generation

Machine Learning Delay Predictor

- **Model Architecture:** Random Forest & Gradient Boosted Regressor
- **Training Library:** Scikit-Learn 1.3.2 (Python)
- **Model File:** transport_model.pkl (Serialized Joblib)
- **Dataset Scale:** 5,000+ historical trip records (CSV)
- **Reported Accuracy:** 97.8% Delay Estimation Accuracy
- **Output Unit:** Expected delay in minutes (+/- 2 mins)
- **Feature Weights:** Rush hour > Weather > Weekend > Ghats
- **Fallback Engine:** Client-side mathematical ML inference fallback

Feature Engineering & Predictor Parameters

The Machine Learning model analyzes four distinct feature vectors in real time:

1. Temporal & Rush Hour Factor:
   - Morning Rush Window (08:00 - 10:30 AM): +18% to +35% congestion multiplier.
   - Evening Rush Window (17:00 - 20:30 PM): +22% to +40% congestion multiplier.
   - Off-Peak Night Transit (22:00 - 05:00 AM): High-speed cruising at 75-85 km/h.

2. Weekend & Holiday Surge Index:
   - Friday evening outbound travel from Chennai, Coimbatore & Bangalore corridors.
   - Sunday evening inbound return surges across Madurai, Trichy & Salem tolls.

3. Satellite Weather & Road Surface Condition:
   - Clear Sky: Baseline 0 minute penalty; optimum asphalt traction.
   - Dense Monsoon Rain: +15 to +25 minutes delay; wet asphalt caution.
   - Thick Highway Fog: +20 to +40 minutes delay; visibility under 100 meters.
   - Severe Storm: +45 to +90 minutes delay; potential road diversion safety hold.

4. Ghat Road Topography Index:
   - Ooty / Nilgiris (36 hairpin bends) & Kodaikanal (Ghat road elevation): Speed restricted to 25 km/h.

Security, Database Engine & System Compliance

Authentication, Storage & Privacy

Authentication & Security Architecture

- **Credential Verification:** Email/Phone/Username/Email & Password Validation
- **Wrong Password Defense:** Immediate block, error banner & shake animation
- **Live Availability Check:** Real-time username & email check during registration
- **Password Reset Flow:** 2-step verified reset via registered email
- **Privacy Enforcement:** Zero public exposure of database memory keys
- **Access Gate:** Full UI blur lock until authenticated session
- **Password Masking:** Native password type with toggle eye button
- **Input Sanitization:** escapeHtml() to protect against XSS injections

Client Database & Storage Engine

- **Database Engine:** W3C HTML5 localStorage (Client-Side DB)
- **User Directory Key:** tn_registered_users (Serialized JSON)
- **Active Session Key:** tn_user (User profile & district metadata)
- **Theme Key:** tn_theme (Persistent active color palette)
- **Tickets Database:** tn_tickets (Booked digital E-tickets & QR)
- **Storage Quota:** Up to 5 MB per origin (sufficient for 10,000+ use
- **Latency:** 0 ms latency (instant in-memory read/write)
- **Offline Capability:** Fully functional without active internet connecti

System Verification, Browser Compatibility & Summary

[Browser Compatibility Matrix]

- Google Chrome: Version 90+ (Full Support for Web Audio API, Leaflet, CSS Grid, Chart.js)
- Microsoft Edge: Version 90+ (Full Chromium compatibility)
- Mozilla Firefox: Version 88+ (Full CSS backdrop-filter & Flexbox support)
- Apple Safari: Version 14+ (Full Web Audio & localStorage support on macOS and iOS)
- Mobile Browsers: Chrome Mobile, Safari iOS, Samsung Internet (Responsive bottom-nav view)

[Technology Stack Summary Table]

- Frontend: Vanilla JavaScript (ES6+), HTML5, CSS3 Glassmorphism, Leaflet.js, Chart.js, FontAwesome 6
- Backend: Python 3.10+, Flask 3.1.3, Werkzeug WSGI server, Scikit-Learn 1.3.2, Pandas, NumPy
- Audio Engine: HTML5 Web Audio API Dual-Oscillator Sawtooth Air Horn (370 Hz / 466 Hz)
- Routing & ML: Dijkstra Shortest Path Highway Algorithm + Random Forest Regression Delay Predictor
- Database: W3C LocalStorage JSON Store (Users, Permissions, Sessions, Tickets, GPS Telemetry)